# Presentation script:

**SLIDE 1 – Title Slide**

**Presenter 1**

Assalamu Alaikum and good morning, respected sir.

This is Najma Akter Lopa, and my teammates are Sultan Mahamud Opu and Sourav Ray.

Today, we are going to present our project titled "Emphasizing Visual Recognition of the American Sign Language Alphabet."

In this project, we focus on reducing the communication gap between the deaf community and hearing individuals using deep learning techniques.

**SLIDE 2 – Outline**

🎙️ **Presenter 1**

This slide shows the outline of our presentation.

First, we will discuss the problem domain in the introduction part.
Then we will discuss the problem statement and objectives.
After that, we will review related work and explain our methodology, including data preprocessing.

Next, we will present the results and real-time decision support system, followed by the conclusion and future work.

**SLIDE 3 – Introduction**

🎙️ **Presenter 1**

Sign language is a visual form of communication that uses hand gestures, facial expressions, and body movements.

According to the World Health Organization, around 1.5 billion people worldwide suffer from hearing loss, and 430 million of them require immediate support.
By 2050, this number may increase to 700 million.

In the United States, about 95% of deaf children are born to hearing parents, most of whom do not know about sign language.

This creates serious communication barriers in education, healthcare, and employment.

Therefore, developing an automated ASL recognition system is extremely important.

**SLIDE 4 – Problem Statement**

🎙️ **Presenter 1**

Most hearing people do not understand sign language, which creates a communication gap.

Relying on human interpreters is costly and not practical in all situations.

Since many hand gestures look similar, building an accurate automated system is challenging, so a strong sign language recognition system is needed.

**SLIDE 5 – Objectives**

🎙️ **Presenter 1**

Our research has three main objectives:

First, to develop a high-accuracy deep learning classifier for 24 static ASL alphabet letters excluding ‘j’ and ‘z’.

Second, to improve model robustness using advanced data augmentation techniques, so the system works well under different lighting conditions, angles, and backgrounds.

Third, to deploy an interpretable real-time Decision Support System, providing visual feedback for better usability.

**SLIDE 6 – Related Work**

🎙️ **Presenter 2**

Thank you, Lopa. At this point, I will handle the rest of the presentation.

We reviewed several existing studies on ASL recognition using both traditional image processing and deep learning techniques.
Early approaches use methods like **HSV color models and edge detection**,but their accuracy is very **low**.

Later, **CNN-based and transfer learning models** significantly improved performance but it struggle to distinguish **similar-looking gestures** like **M, N, and S**.

Most existing systems **don't have any real-time decision support**, they also use **small or limited datasets**.

In contrast our model achieves superior accuracy with a real time dss that clearly fill those research gap.

🔹 **SLIDE 7 – Data Description**

🎙️ **Presenter 2**

Now, let's talk about our dataset.

For our experiments, we used the ASL Alphabet dataset that was collected from Kaggle.
The dataset contains 2,427 RGB images in JPG format, with an average of 101 images per class, which means the classes are well balanced.

All images were resized to 224 × 224 pixels, making them suitable for transfer learning models.

🔹 **SLIDE 8 – Core Preprocessing (1/4)**

🎙️ **Presenter 2**

Here start our data preprocessing face.

Data preprocessing is a very important step in our system. Let me briefly explain it.

AT First,We  load all images and resized them to a fixed resolution of 128 × 128 pixels.
This ensures consistent input for the network and also reduces computational complexity.

🔹 **SLIDE 9 – Core Preprocessing (2/4)**

🎙️ **Presenter 2**

Our main preprocessing starts with geometric data augmentation. Specifically, here we applied:

- **Random rotation from** –12° to +12° to make the model more robust to different hand angles.
- **We aslo use 4x**  Zoom in and out to simulates changes in distance from the camera.
- **We also use Random shifts like** Horizontal and vertical shifts that help the model to handle variations in hand position.
- **There also Random use shearing transformation.**

These augmentations help the model work well with different camera positions and angles.

🔹 **SLIDE 10 – Core Preprocessing (3/4)**

🎙️ **Presenter 2**

In addition, we applied photometric data augmentation techniques to make the model more robust to different visual conditions. Specifically, we used:

- **Brightness & Contrast Adjustment** to handle different lighting conditions.
- Adding random **Gaussian Noise** to simulate sensor or environmental variations.
- **Gaussian Blur** is applied to reduce sensitivity to minor focus changes.
- **Motion Blur:** Simulating hand movement to improve robustness to motion.
- **Color Adjustments:** Randomly modifying hue, saturation, or color balance to account for different skin tones and background variation.

🔹 **SLIDE 11 – Core Preprocessing (4/4)**

🎙️ **Presenter 2**

This slide shows examples of original and augmented images.

- In Original images there is no modification.
- In **Augmented 1, 2, 3, 11:** Random black patches are added to make the model robust to partial occlusion.
- In Augmented 4, 8 and 10 Color Adjustments are use to simulate changes in lighting and skin tone.
- In **Augmented 5, 6 and 9:** Gaussian noise adds random pixel to mimic sensor or environmental noise.
- Lastly in Augmented 7  Random Occlusion are use.

🔹 **SLIDE 12 – Methodology**

🎙️ **Presenter 3**

**Now let's discuss** our proposed methodology,

at fist we collecte our raw data from kaggle.Then we perform data analysis like hand detection, resolution check, background inspection, noise detection etc.

Next comes the preprocessing stage, which I explained before.

After preprocessing, we split our dataset into two part, 70% for training, and 30% for testing.

Then comes the most important part, model training. Here we use several deep learning models like MobileNetV2, MobileNetv3, vision transformer, restnet50 etc.

We further optimized the models using **grid search hyperparameter tuning**.
Finally, we evaluate our model by using some evaluation metrics like accuracy, f1 score, precision, recall etc.
In addition, we developed a **real-time interpretable Decision Support System (DSS)** to provide instant support and predictions.

*Now, I will hand it over to my next teammate, Sourav Ray, to explain the rest of the presentation.*

🔹 **SLIDE 13 – Results (Optimized Hyperparameters)**

🎙️ **Presenter 3**

After optimization, we observed improved accuracy and stable training behavior across models.

These results confirm the effectiveness of our preprocessing and model selection.

🔹 **SLIDE 14 – Results**

🎙️ **Presenter 3**

The **Vision Transformer model** achieved the highest accuracy of **94.10%**.

EfficientNetB0 and MobileNetV3 also showed strong performance with a good balance between accuracy and efficiency.

🔹 **SLIDE 15 – Real-Time DSS**

🎙️ **Presenter 3**

Our system supports **real-time decision support**, allowing live ASL alphabet recognition.

This makes the solution suitable for assistive technologies and educational applications.

🔹 **SLIDE 16 – Conclusion**

🎙️ **Presenter 3**

In conclusion, our project demonstrates that **advanced vision models**, especially Vision Transformers, can effectively recognize ASL alphabets.

The system achieves high accuracy and robustness, making it suitable for real-world use.

**SLIDE 18 – References**

🎙️ **Presenter 3**

The references listed here include research papers, datasets, and reports that supported our work.

🔹 **SLIDE 19 – Thank You**

🎙️ **Presenter 3**

**Thank you for your attention.**